

01st-10-2026

International Cybersecurity Of Digital Forensic Academy (ICDFA)

PROGRAMME: Advance Critical Infrastructure Security (ACIS)

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Course: ACICP301-INTRO TO CRITICAL INFRASTRUCTURE
FOR IT LEARNERS

Title: EXPLORING ICS PACKET CAPTURES

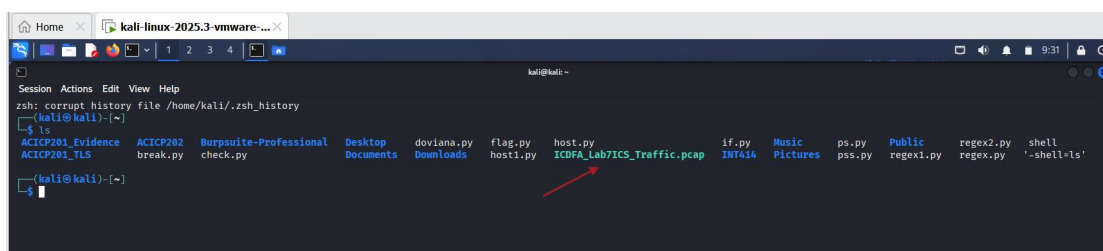
Assignment: Week 1-Lab 2 Report
C11/26/ACIS/17359

Instructor : Aminu Idris

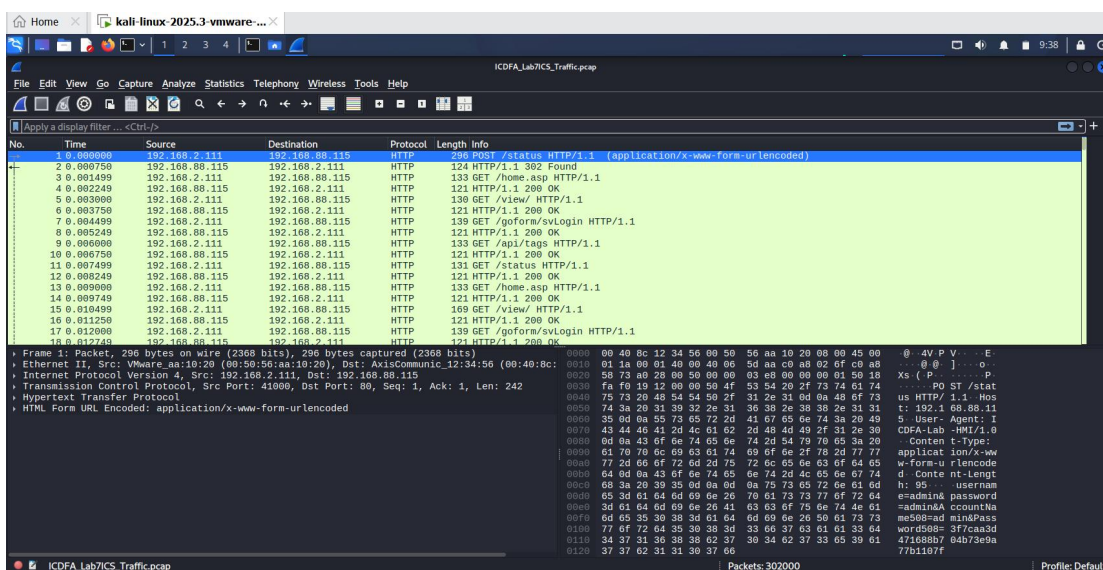
Student Name	Aminu Ibrahim Jalo
Registration Number	C11/26/ACIS/17359
Signature	
Date	01 st -10-2026

Procedure

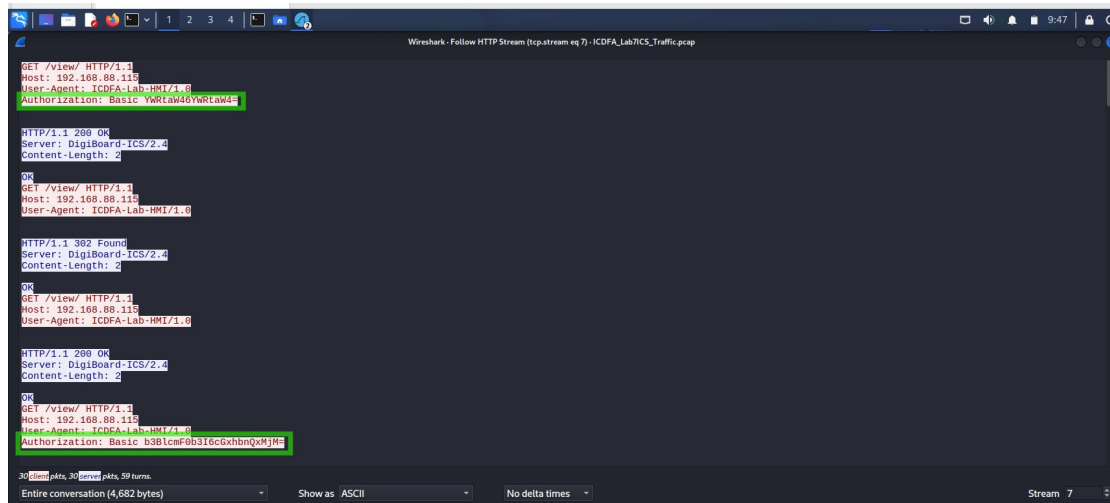
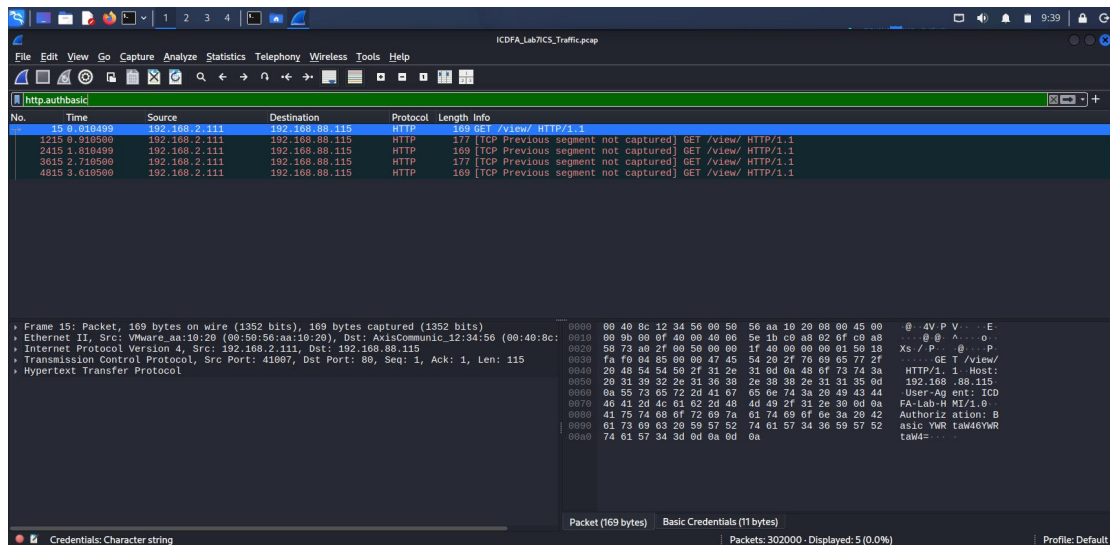
Step 1. Start the ICDFA analyst VM in an isolated environment. This lab uses instructor-provided packet captures only; no scanning, exploitation or credential reuse is required.



Step 2. Open Wireshark and load the first instructor-provided ICS capture (for the teaching dataset this may be 4SICS-GeekLounge-151021.pcap).



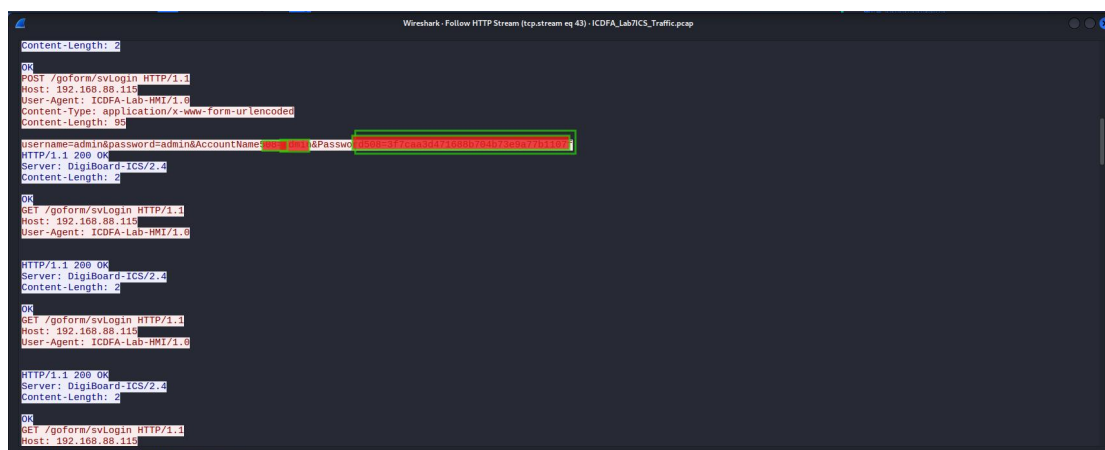
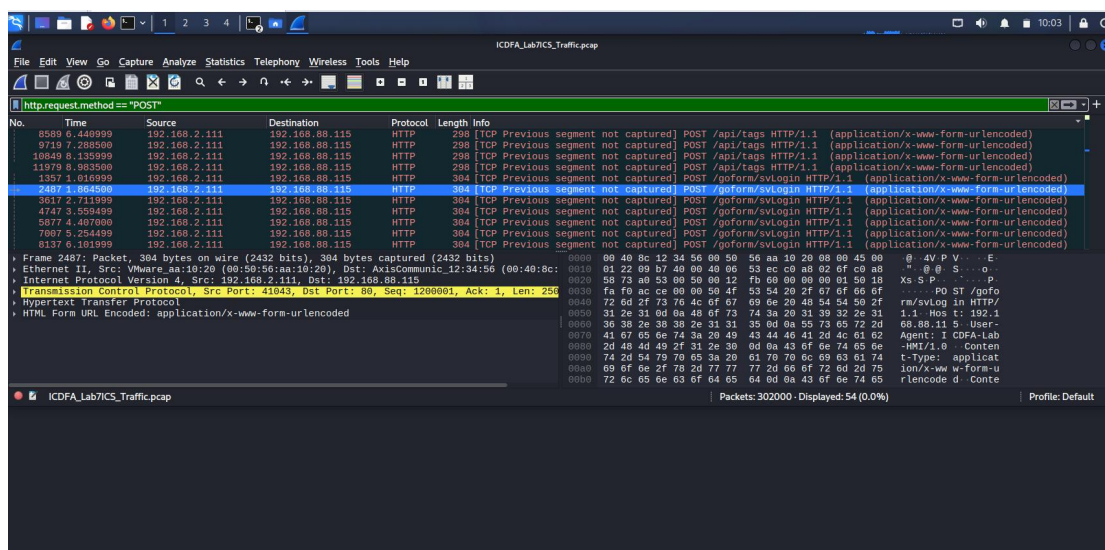
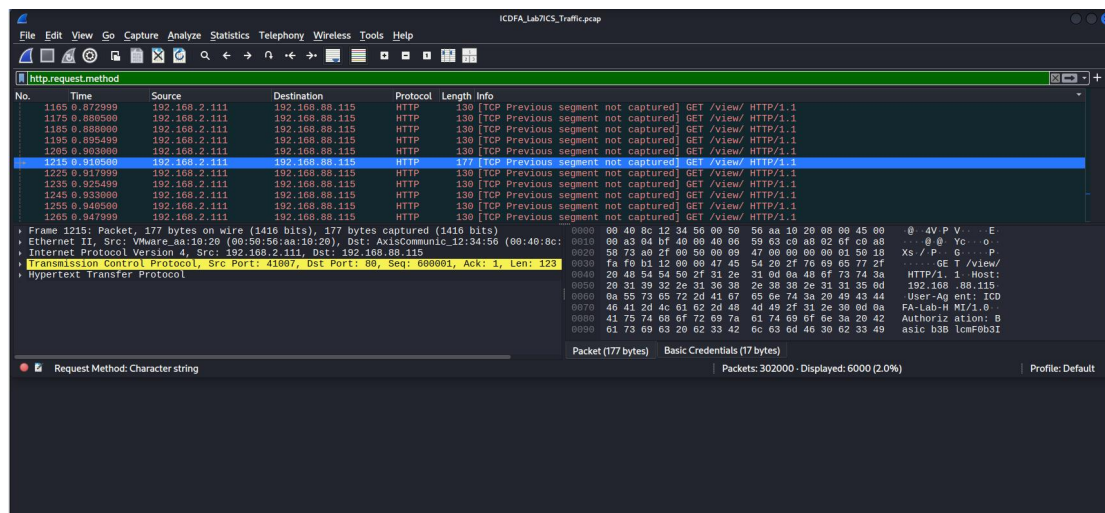
Step 3. Apply the filter **http.authbasic**. Inspect the HTTP Authorization field and determine whether Basic authentication material is present. If you decode an encoded value for learning purposes, do so only within the provided capture and redact the recovered credential from submitted screenshots.



```
echo 'YWRtaW46YWRtaW4=' | base64 -d | sed 's/:.*/:*****/'; echo
echo 'b3BlcmF0b3I6cGxhbnQxMjM=' | base64 -d | sed 's/:.*/:*****/'; echo
echo
```



Step 4. Clear the filter and apply **http.request.method**. Refine to **http.request.method == "POST"** and inspect form submissions, request paths and response status codes. Record security-relevant observations without attempting to log in to any identified device.

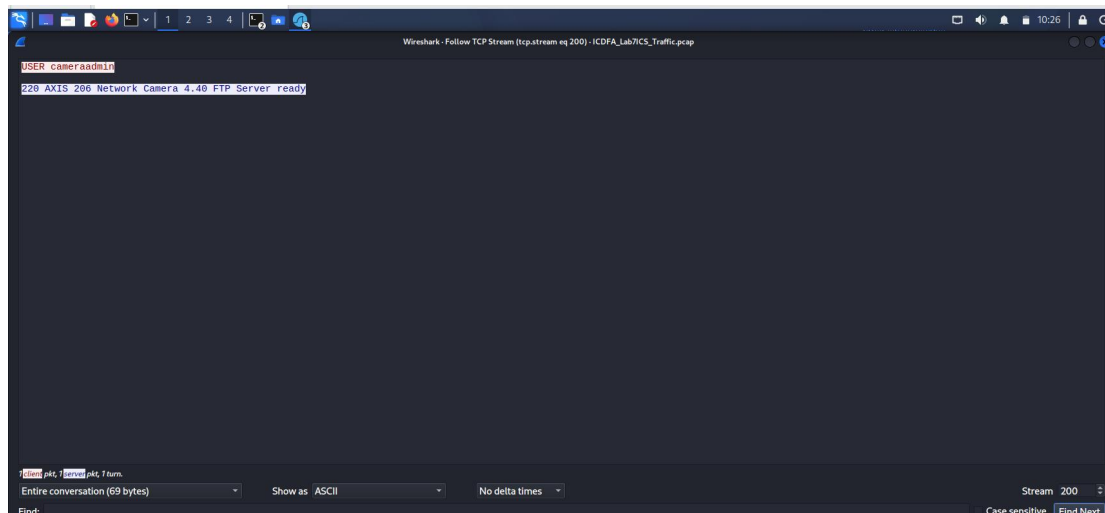
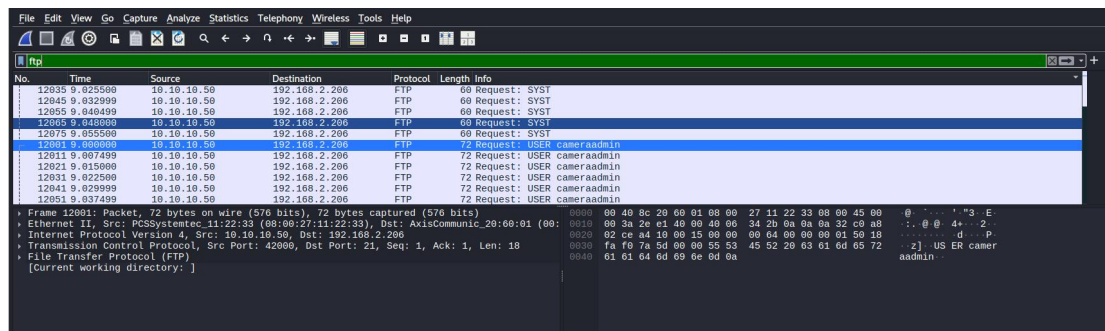


Observation (HTTP Stream, POST /goform/svLogin):

Following the HTTP stream for the **POST** to /goform/svLogin shows the HMI client (ICDFA-Lab-HMI/1.0) sending a login form to 192.168.88.115 over unencrypted HTTP. The form body is URL-encoded and carries four fields: username, password, AccountName508 and Password508 (values redacted). The password field appears in cleartext, and Password508 is a 32-character hexadecimal value, consistent with a

hash, sent alongside it. The server (DigiBoard-ICS/2.4) replied 200 OK with a 2-byte OK body, so the capture doesn't show whether the login succeeded. Because everything is readable on the wire and the hash adds no confidentiality, any host able to capture this traffic can recover or replay the credentials. No access to the device was attempted.

Step 5. Apply ftp and inspect the FTP session metadata. Record any product/vendor or software-version information visible in the capture.

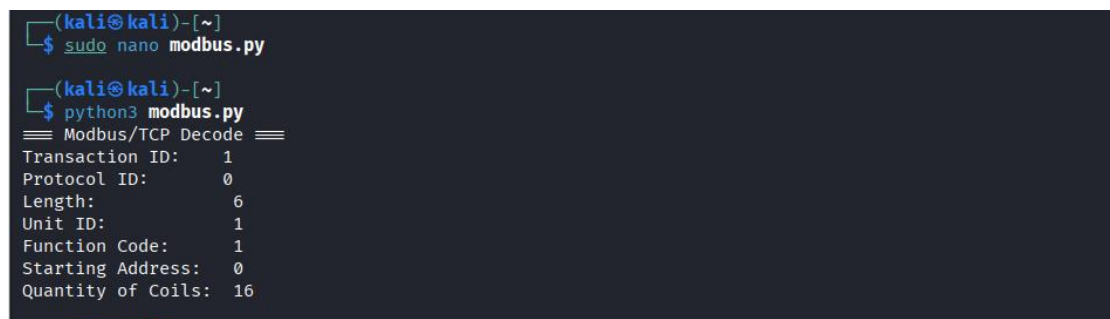
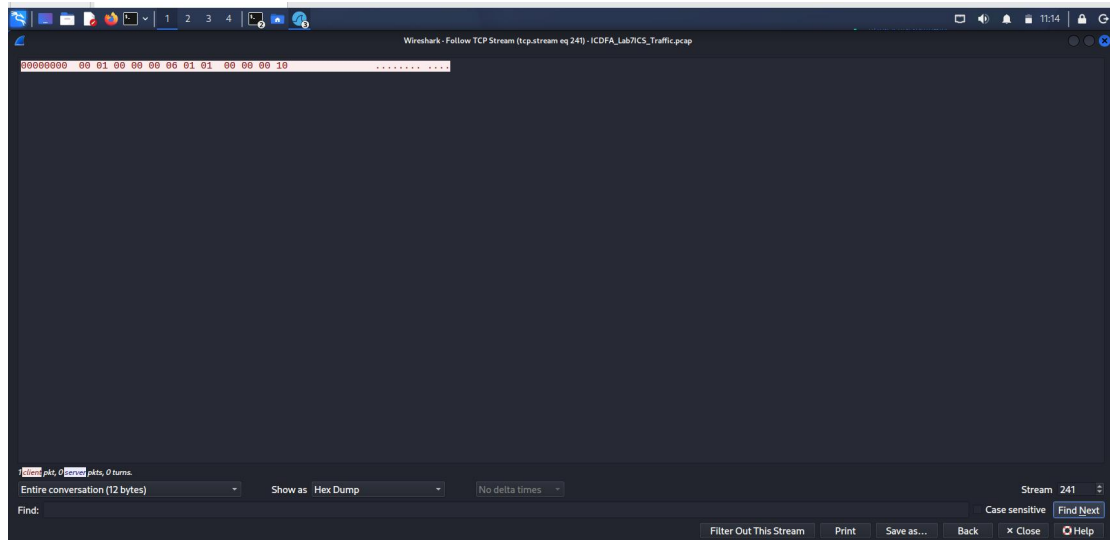
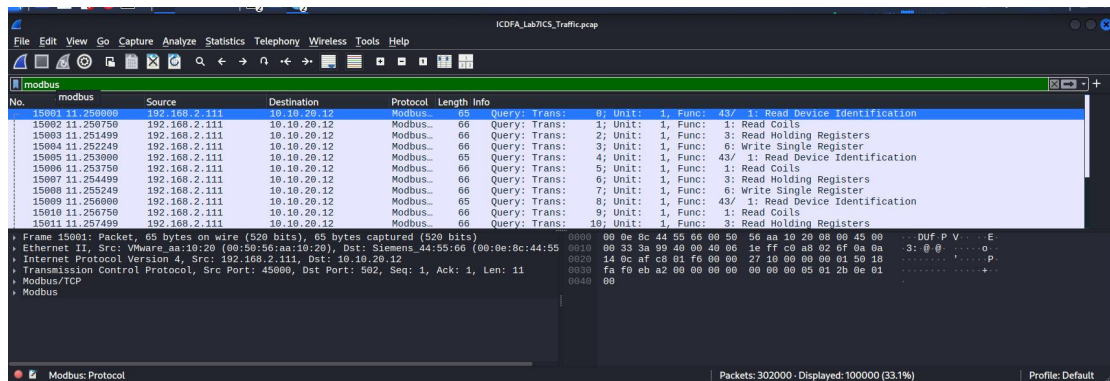


Observation:

The FTP session between client IP and camera IP is unencrypted. The server banner discloses the vendor, model and software version (AXIS 206 Network Camera 4.40). The client authenticates with the USER command using an administrative-style account name (redacted), and the credentials travel in cleartext. Anyone able to capture this traffic can read the account details, and the exposed model and version let an observer look up known weaknesses for that device. The presence of a network camera on this segment also shows that non-control devices are visible alongside ICS traffic. No connection to the device was attempted.

Step 6. Apply modbus and identify at least two Modbus function codes. If the second teaching capture is supplied, open it and compare the volume/types of Modbus traffic.

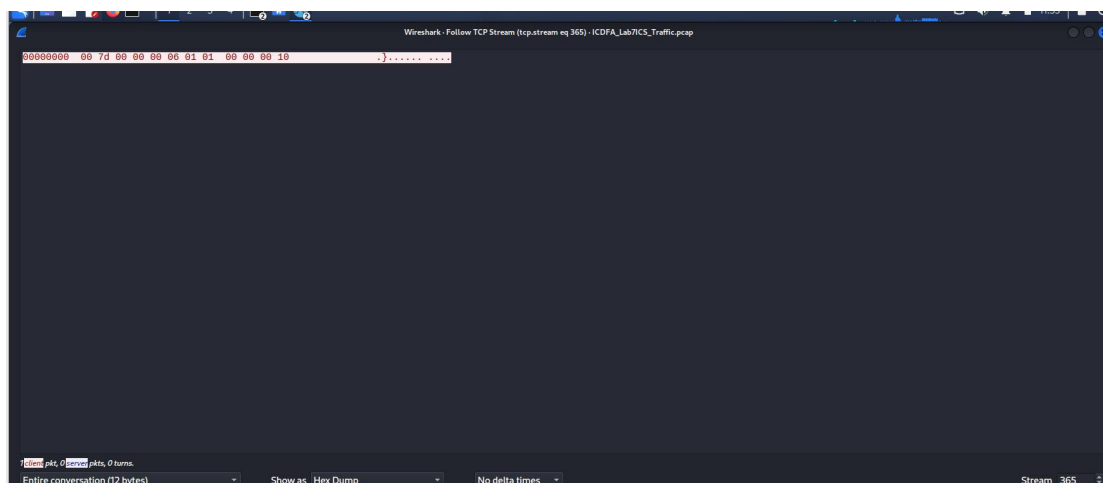
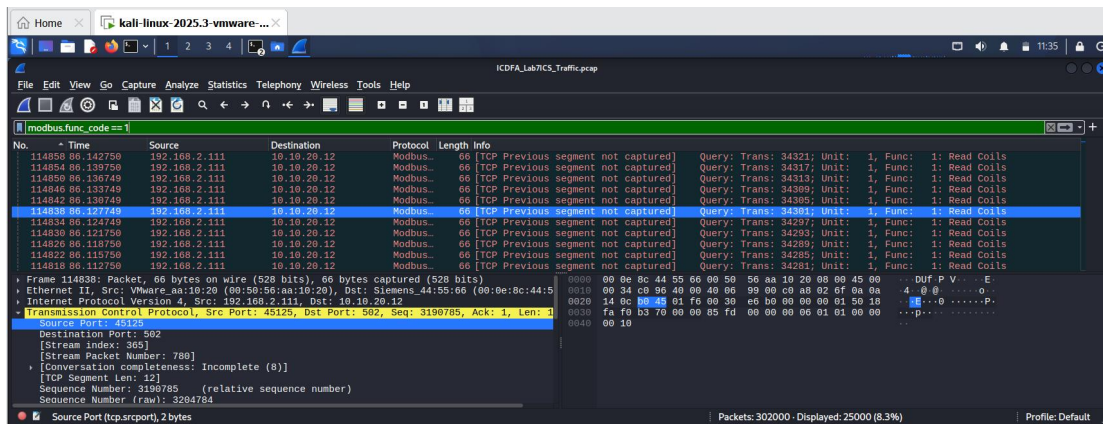
Filter: modbus



Observation:

Packet 15001 – Modbus/TCP Read Coils Request: The packet contains a Modbus/TCP request with Unit ID 1 and Function Code 01. The client requests the status of 16 coils beginning at address 0.

Filter: modbus.func_code == 1



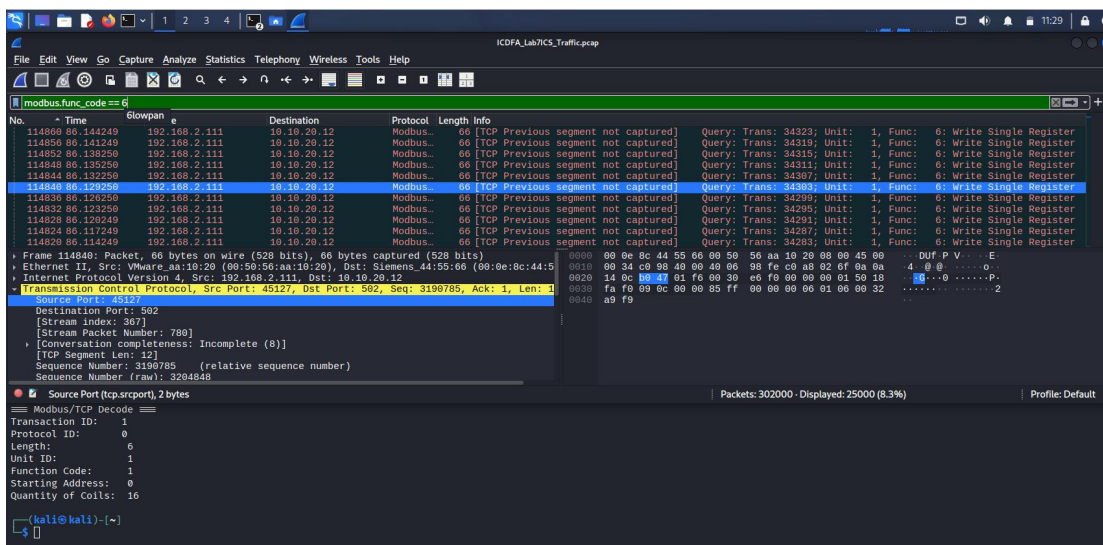
output:007d00000006010100000010

Observation:

Function Code 01 - Read Coils

The packet requests the status of **16 coils starting from address 0** on Unit ID 1. This is typically used to read binary on/off states from an industrial device.

Filtter: modbus.func_code == 6



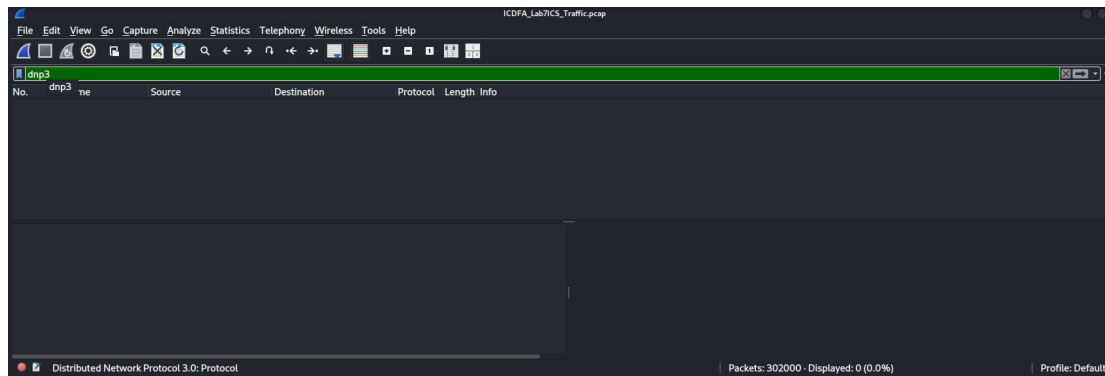
Output: 007f00000006010600320379

Observation:

Function Code 06 – Write Single Register

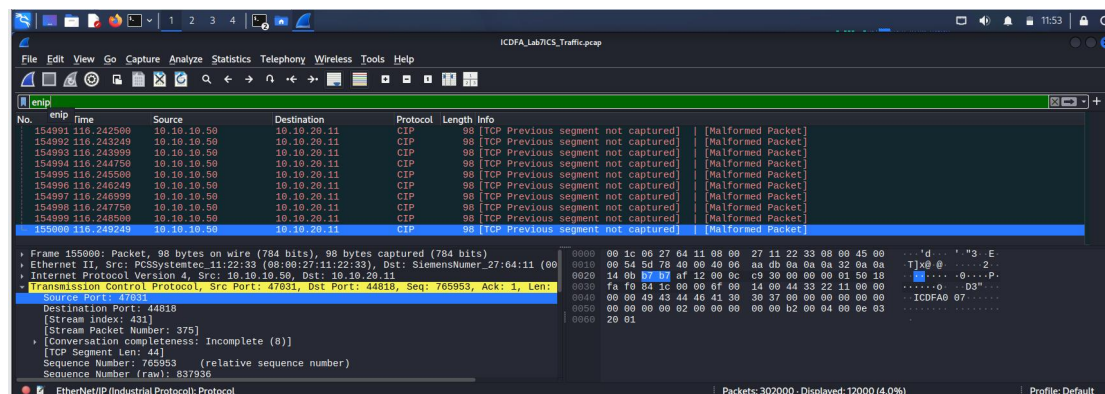
The packet writes the value 889 (0x0379) to register 50 on Unit ID 1. This demonstrates that the Modbus traffic includes a write operation capable of changing a register value.

Step 7. Apply dnp3 and identify a DNP3 conversation. Record source/destination systems and explain why DNP3 is common in process automation/SCADA environments.



DNP3: No DNP3 traffic was identified in the supplied capture using the dnp3 filter. Therefore, no DNP3 source/destination conversation was available for analysis. And DNP3 is commonly used in SCADA and process-automation environments for communication between control-center systems and remote field devices.

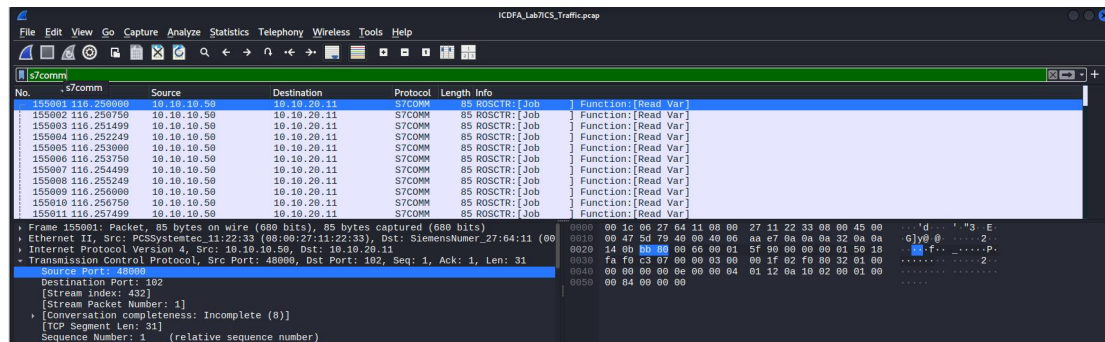
Step 8. Apply enip and inspect available EtherNet/IP/CIP traffic. Record the key header/protocol details visible in the capture.



EtherNet/IP Observation: The packet uses EtherNet/IP SendRRData (0x006F) with a 20-byte command-specific data section and session handle 0x11223344. The

encapsulated data contains CIP-related information, demonstrating EtherNet/IP communication used for industrial device and control-system messaging.

Step 9. Apply s7comm and inspect an S7 communication exchange. Record source/destination information and any Read Var or comparable operation visible.



Packate 155001-TCP

S7comm Observation: An S7comm Read Var operation was observed in the capture. The packet uses TPKT/COTP transport and contains an S7comm request for variable data, demonstrating communication between an S7 client and an industrial automation device.

Step 10. Complete the Protocol Evidence Matrix for HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7.

DONE

Step 11. Write a security assessment identifying at least five observations such as cleartext/encoded credentials, legacy protocols, exposed device metadata, insufficient confidentiality or excessive network visibility.

Security Assessment

The analysis of the provided ICS packet capture identified several security concerns associated with the use of legacy industrial protocols and the visibility of sensitive communication information on the network.

Cleartext or encoded credentials: HTTP Basic Authentication traffic was observed. Although credentials may appear encoded using Base64, Base64 does not provide encryption or confidentiality. An attacker with access to the network traffic could potentially recover the original credentials.

Use of legacy industrial protocols: Protocols such as Modbus/TCP and S7comm were identified in the capture. These protocols were designed primarily for reliable industrial communication and generally provide limited or no built-in protection against interception or manipulation. Modbus traffic also demonstrated read and write operations, including Read Coils and Write Single Register functions.

Exposed device and protocol metadata: EtherNet/IP, S7comm, Modbus and other protocol information can reveal details about industrial communication systems. Function codes, device addresses, session information, protocol types and communication patterns can help an observer understand the architecture and operations of an ICS environment.

Insufficient confidentiality: Several industrial protocols transmit operational information without strong native encryption. Passive monitoring can therefore expose information about control requests, device communication and process-related activity.

Excessive network visibility: The capture demonstrates that multiple industrial and IT protocols can be observed from network traffic. If industrial networks are insufficiently segmented, unauthorized users who gain network access may be able to observe communications between controllers, servers and field devices.

Write-capable industrial communication: The Modbus traffic included a Function Code 06 (Write Single Register) operation. The observed packet wrote the value 889 to register address 50 on Unit ID 1. This demonstrates that industrial traffic may contain commands capable of changing device or process-related values and therefore requires appropriate network and access controls.

Information exposure through protocol characteristics: Protocol-specific information such as EtherNet/IP SendRRData, Modbus function codes and S7comm Read Var operations provides useful context about how devices communicate. Such information can increase an adversary's understanding of the industrial environment if the traffic is accessible.

Recommended Compensating Controls

Where replacement of legacy protocols is not immediately practical, organizations should use network segmentation and industrial DMZs, restrict access using firewalls and allowlists, apply strong authentication where supported, use encrypted tunnels or secure gateways where appropriate, monitor ICS traffic for anomalous activity, minimize unnecessary network exposure, and maintain detailed logging and incident-response procedures. Passive monitoring and network intrusion detection can provide additional visibility without directly interacting with industrial devices.

Overall Assessment

The packet capture demonstrates that legacy ICS protocols can expose authentication material, device information and operational communication to anyone with sufficient network visibility. Strong segmentation, access control, monitoring and compensating security controls are therefore important for protecting industrial environments while legacy protocols remain in use.

Step 12. Save your Wireshark workspace/screenshots and submit the report. Do not use any discovered credentials, addresses or device information outside the provided training material.

Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Response: Base64 is an encoding method, not encryption. It can be easily decoded back into the original username and password. Therefore, HTTP Basic Authentication should be protected with HTTPS/TLS to provide confidentiality.

2. What information can device banners or protocol metadata reveal to an analyst?

Response: They can reveal device types, vendors, software or firmware versions, protocol types, IP addresses, communication roles, services, function codes and other information about the industrial network architecture.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Response: Modbus, DNP3, EtherNet/IP and S7 are industrial protocols designed for communication between PLCs, RTUs, HMIs, SCADA systems and other automation devices. HTTP and FTP are primarily designed for general IT and user-facing web or file-transfer services. ICS protocols commonly carry control, monitoring and process data rather than ordinary user content.

4. Why is passive packet analysis valuable for understanding an ICS environment?

Response: Passive packet analysis allows analysts to observe existing network communication without interacting with or modifying industrial devices. It can reveal protocols, communication relationships, device roles, commands, operational patterns and potential security weaknesses while minimizing the risk of disrupting the ICS environment.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Response: Organizations can use network segmentation, industrial DMZs, firewalls and allowlists, strict access controls, encrypted tunnels or secure gateways, continuous network monitoring, intrusion detection, logging, secure remote access, and regular review of unnecessary network exposure. These controls can reduce the risk while legacy protocols remain in use.